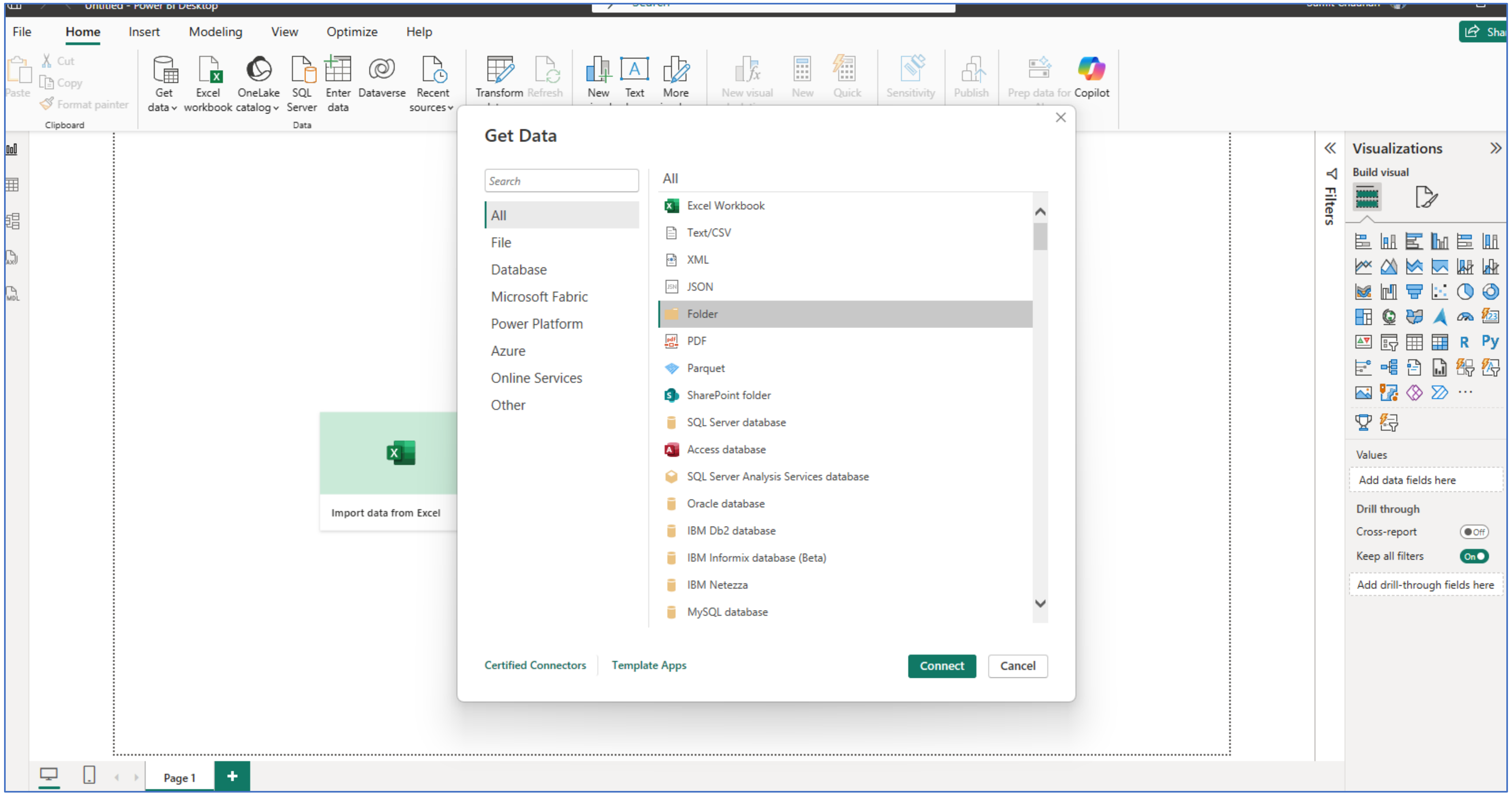
Power Query

Name: - Sumit Chauhan

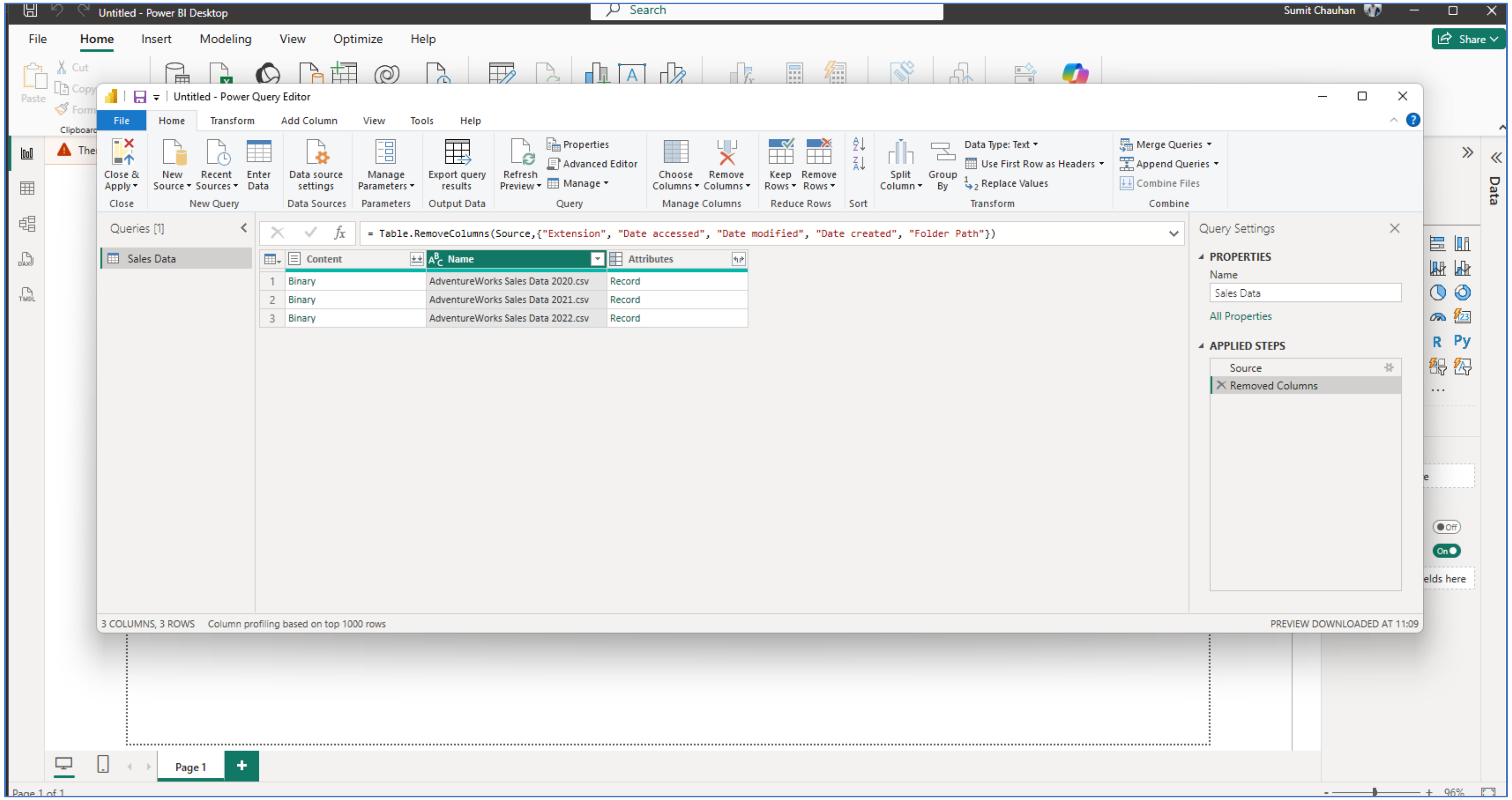
Dept: - Data Engineering

Power Query Assignment

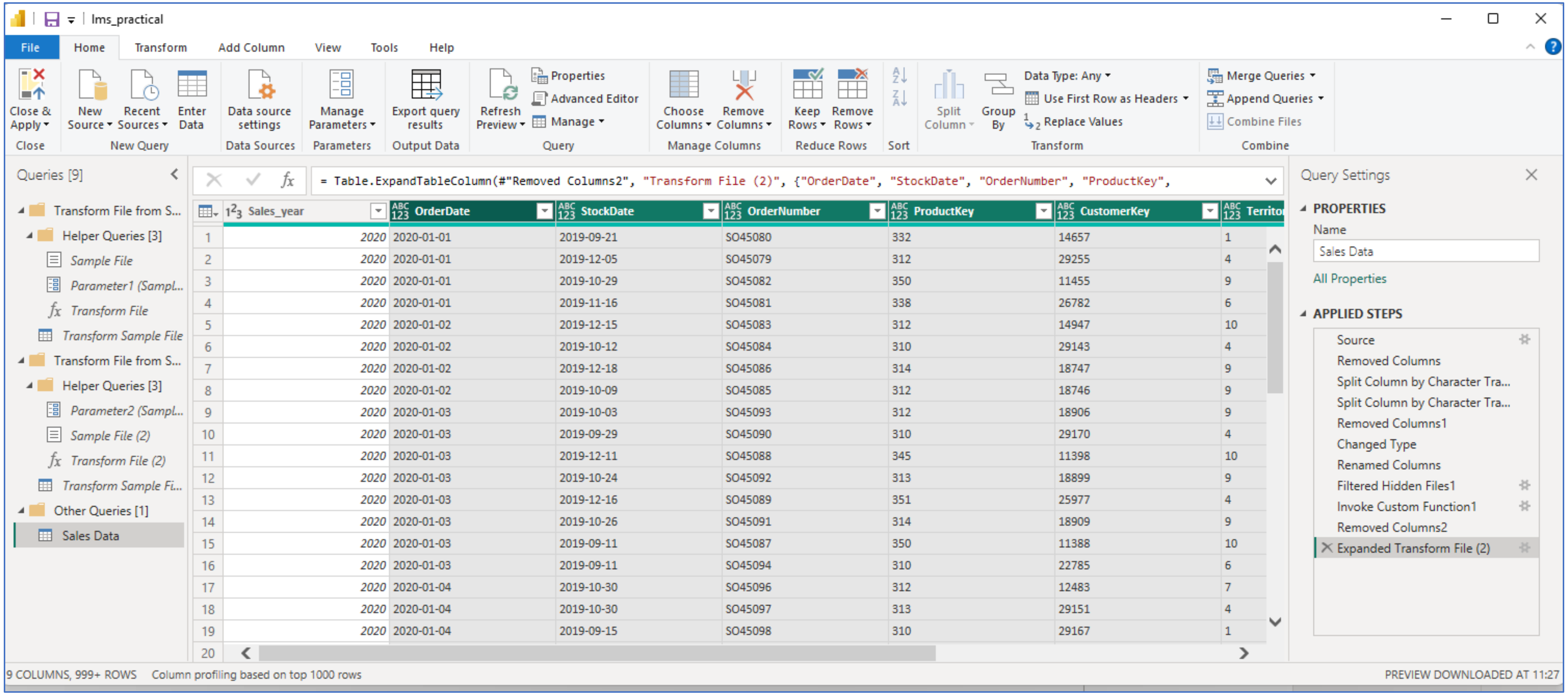
- Create queries to connect to all .csv files in the folder into power query



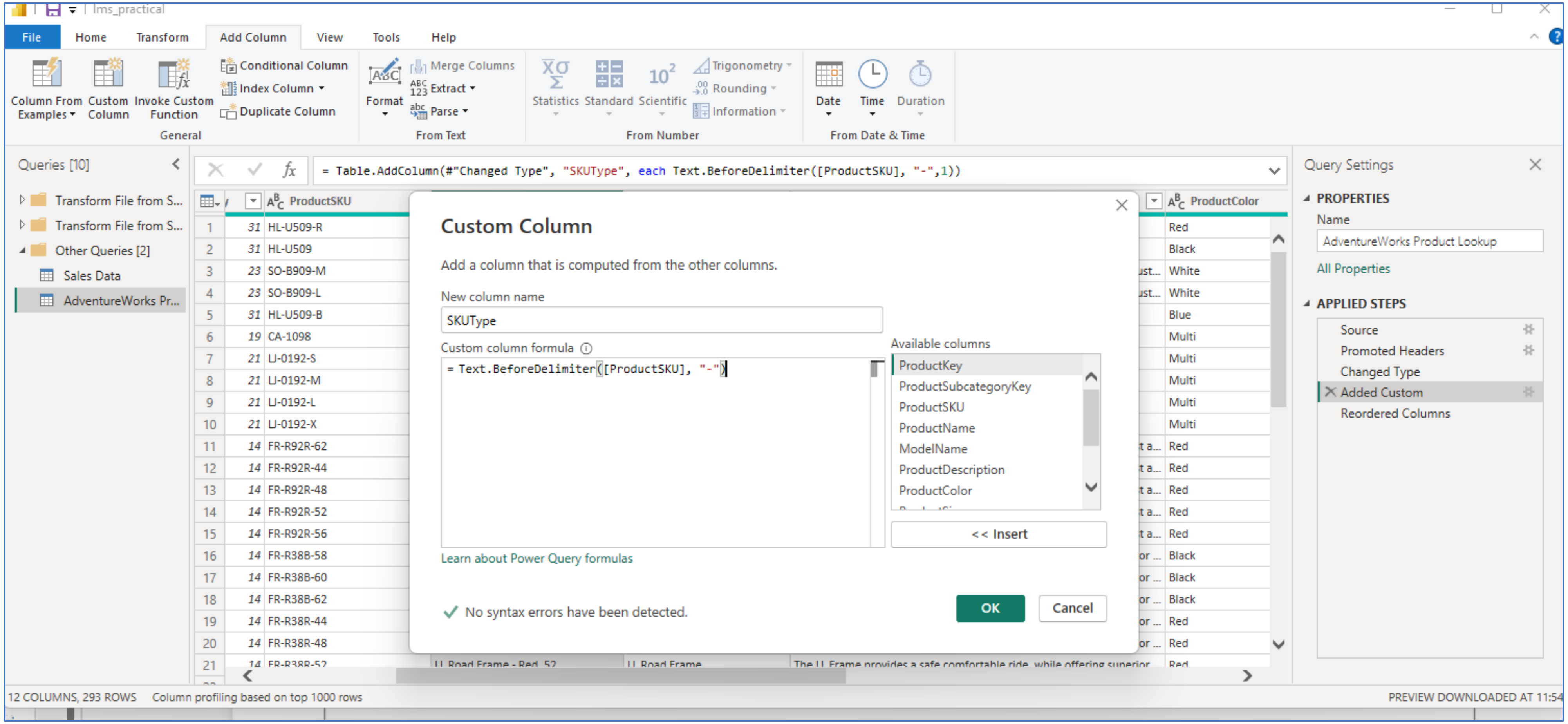
Removing unnecessary columns

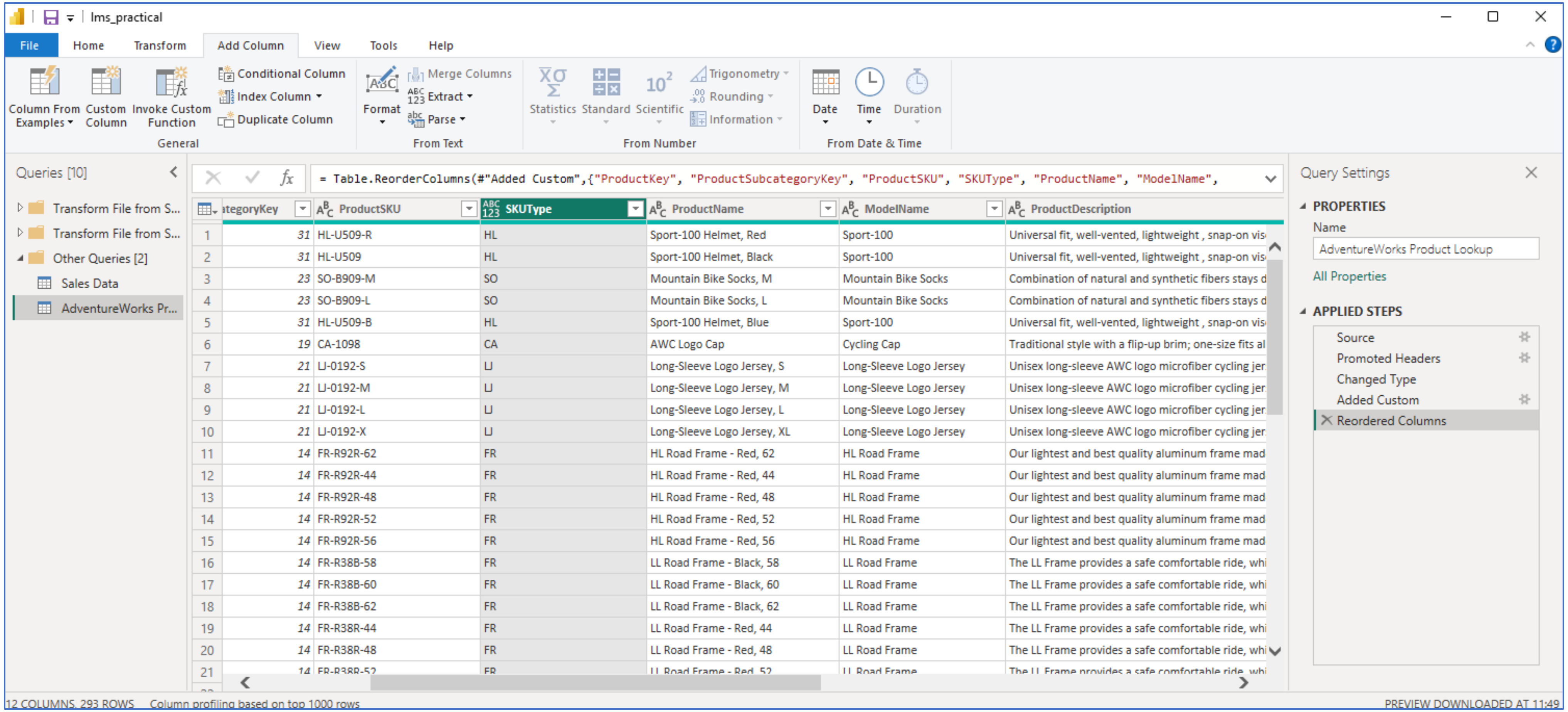


Keeping Sales_year column and merging sales data files

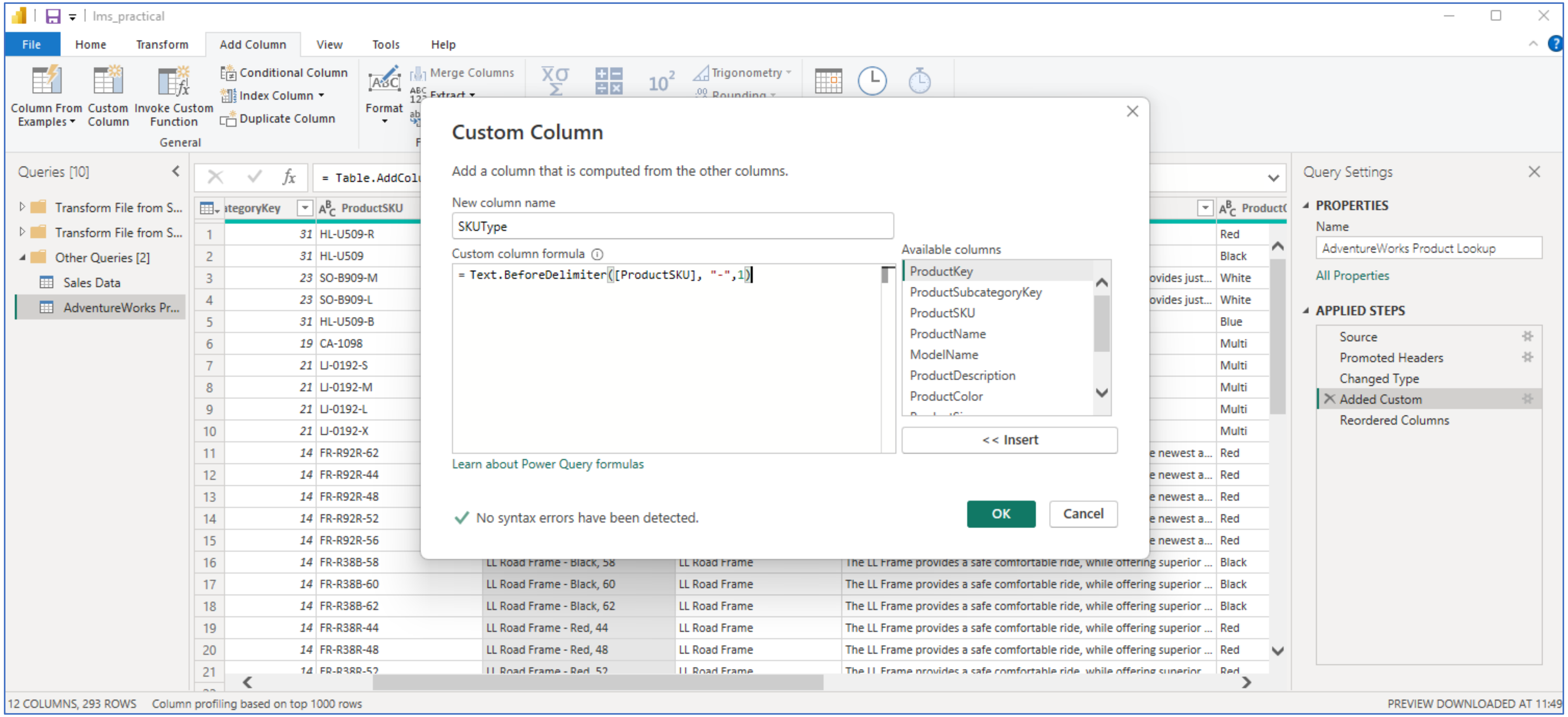


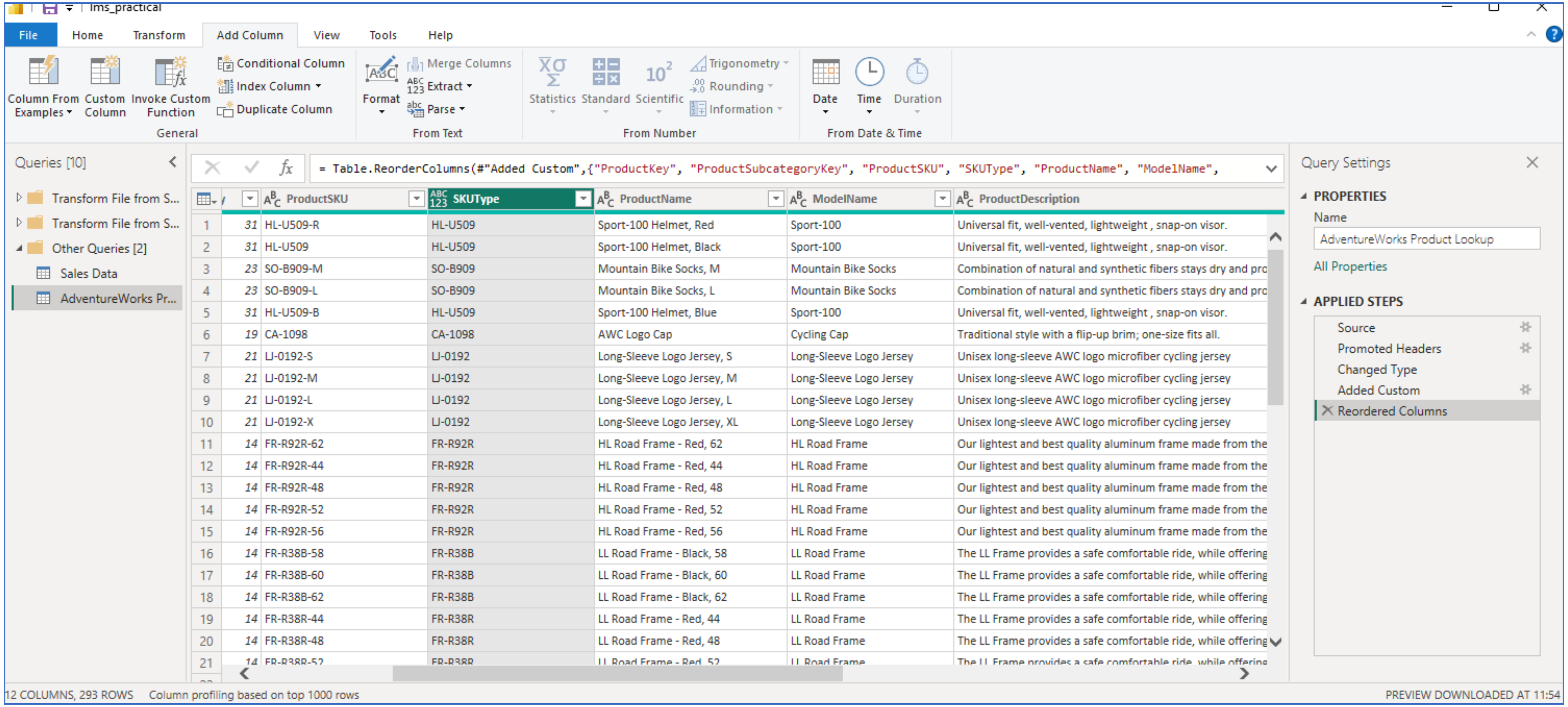
- Add a new column to extract all characters before the dash (""-"") in the Product SKU column, and name it “SKU Type”




lms_practical
File
Home
Transform
Add Column
View
Tools
Help
Column From Examples
Custom Column
Invoke Custom Function
Conditional Column
Index Column
Duplicate Column
General
Format
Merge Columns
Extract
Parse
From Text
Statistics
Standard
Scientific
Trigonometry
Rounding
Information
From Number
Date
Time
Duration
From Date & Time
Queries [10]
Transform File from S...
Transform File from S...
Other Queries [2]
Sales Data
AdventureWorks Pr...
= Table.ReorderColumns(#"Added Custom",{"ProductKey", "ProductSubcategoryKey", "ProductSKU", "SKUType", "ProductName", "ModelName",
tegoryKey
ProductSKU
SKUType
ProductName
ModelName
ProductDescription
1 31 HL-U509-R HL Sport-100 Helmet, Red Sport-100 Universal fit, well-vented, lightweight , snap-on vis
2 31 HL-U509 HL Sport-100 Helmet, Black Sport-100 Universal fit, well-vented, lightweight , snap-on vis
3 23 SO-B909-M SO Mountain Bike Socks, M Mountain Bike Socks Combination of natural and synthetic fibers stays d
4 23 SO-B909-L SO Mountain Bike Socks, L Mountain Bike Socks Combination of natural and synthetic fibers stays d
5 31 HL-U509-B HL Sport-100 Helmet, Blue Sport-100 Universal fit, well-vented, lightweight , snap-on vis
6 19 CA-1098 CA AWC Logo Cap Cycling Cap Traditional style with a flip-up brim; one-size fits al
7 21 LJ-0192-S LJ Long-Sleeve Logo Jersey, S Long-Sleeve Logo Jersey Unisex long-sleeve AWC logo microfiber cycling jer
8 21 LJ-0192-M LJ Long-Sleeve Logo Jersey, M Long-Sleeve Logo Jersey Unisex long-sleeve AWC logo microfiber cycling jer
9 21 LJ-0192-L LJ Long-Sleeve Logo Jersey, L Long-Sleeve Logo Jersey Unisex long-sleeve AWC logo microfiber cycling jer
10 21 LJ-0192-X LJ Long-Sleeve Logo Jersey, XL Long-Sleeve Logo Jersey Unisex long-sleeve AWC logo microfiber cycling jer
11 14 FR-R92R-62 FR HL Road Frame - Red, 62 HL Road Frame Our lightest and best quality aluminum frame mad
12 14 FR-R92R-44 FR HL Road Frame - Red, 44 HL Road Frame Our lightest and best quality aluminum frame mad
13 14 FR-R92R-48 FR HL Road Frame - Red, 48 HL Road Frame Our lightest and best quality aluminum frame mad
14 14 FR-R92R-52 FR HL Road Frame - Red, 52 HL Road Frame Our lightest and best quality aluminum frame mad
15 14 FR-R92R-56 FR HL Road Frame - Red, 56 HL Road Frame Our lightest and best quality aluminum frame mad
16 14 FR-R38B-58 FR LL Road Frame - Black, 58 LL Road Frame The LL Frame provides a safe comfortable ride, whi
17 14 FR-R38B-60 FR LL Road Frame - Black, 60 LL Road Frame The LL Frame provides a safe comfortable ride, whi
18 14 FR-R38B-62 FR LL Road Frame - Black, 62 LL Road Frame The LL Frame provides a safe comfortable ride, whi
19 14 FR-R38R-44 FR LL Road Frame - Red, 44 LL Road Frame The LL Frame provides a safe comfortable ride, whi
20 14 FR-R38R-48 FR LL Road Frame - Red, 48 LL Road Frame The LL Frame provides a safe comfortable ride, whi
Query Settings
PROPERTIES
Name
AdventureWorks Product Lookup
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Added Custom
Reordered Columns
12 COLUMNS, 293 ROWS Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 11:49

- Update the SKU Type calculation above to return all characters before second dash




lms_practical
File
Home
Transform
Add Column
View
Tools
Help
Column From Examples
Custom Column
Invoke Custom Function
Conditional Column
Index Column
Duplicate Column
General
Format
Merge Columns
Extract
Parse
From Text
Statistics
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Scientific
Trigonometry
Rounding
Information
From Number
Date
Time
Duration
From Date & Time
Queries [10]
Transform File from S...
Transform File from S...
Other Queries [2]
Sales Data
AdventureWorks Pr...
= Table.ReorderColumns(#"Added Custom",{"ProductKey", "ProductSubcategoryKey", "ProductSKU", "SKUType", "ProductName", "ModelName",
ProductSKU
SKUType
ProductName
ModelName
ProductDescription
1 31 HL-U509-R HL-U509 Sport-100 Helmet, Red Sport-100 Universal fit, well-vented, lightweight , snap-on visor.
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13 14 FR-R92R-48 FR-R92R HL Road Frame - Red, 48 HL Road Frame Our lightest and best quality aluminum frame made from the
14 14 FR-R92R-52 FR-R92R HL Road Frame - Red, 52 HL Road Frame Our lightest and best quality aluminum frame made from the
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16 14 FR-R38B-58 FR-R38B LL Road Frame - Black, 58 LL Road Frame The LL Frame provides a safe comfortable ride, while offering
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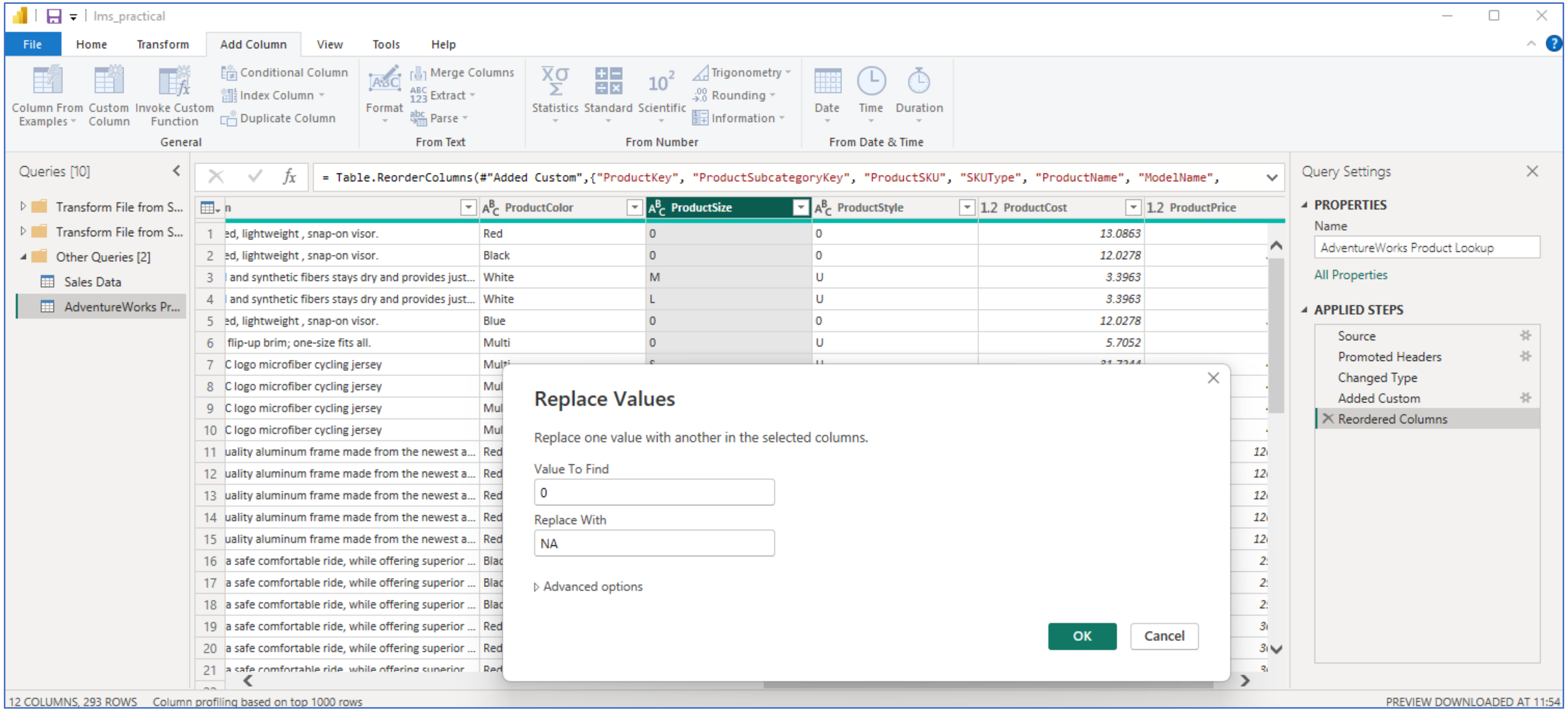
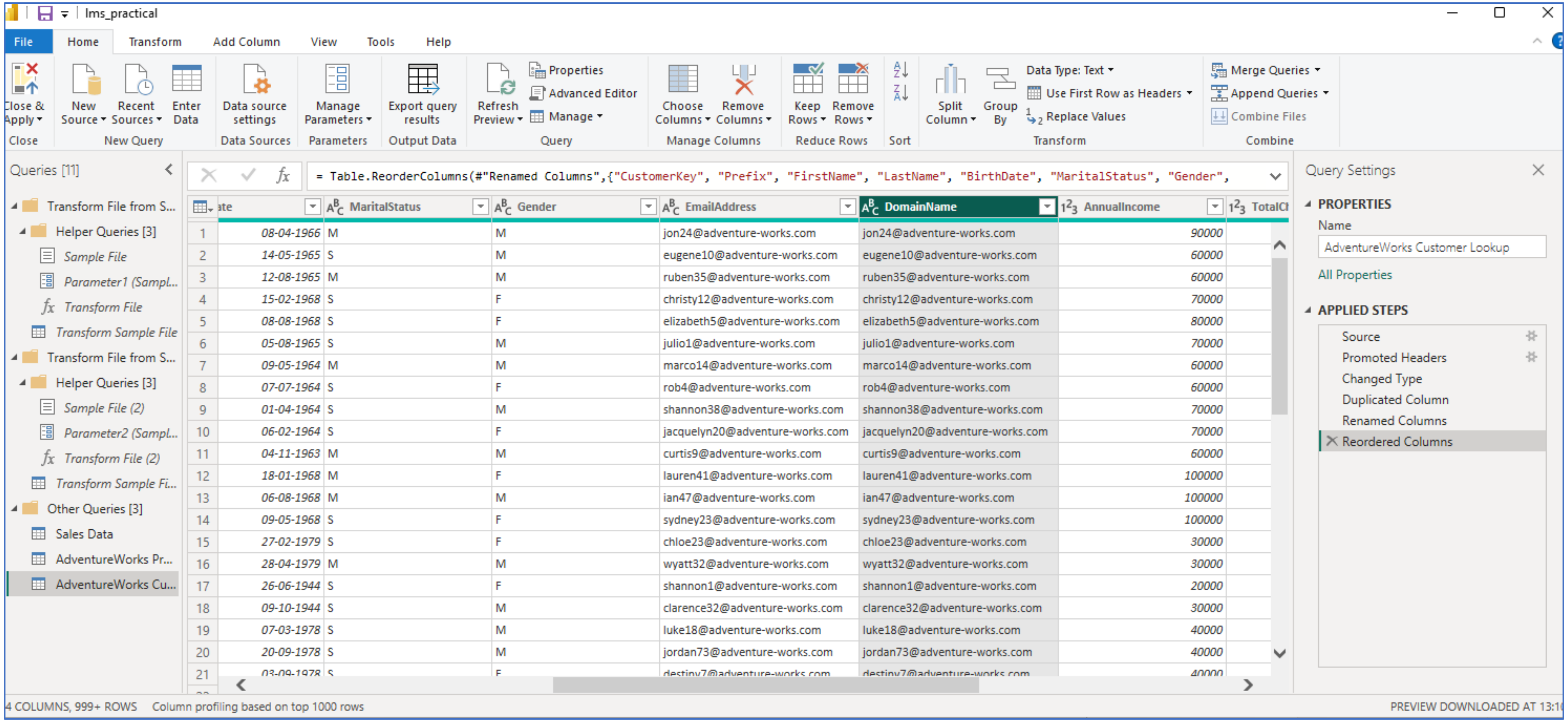
18 14 FR-R38B-62 FR-R38B LL Road Frame - Black, 62 LL Road Frame The LL Frame provides a safe comfortable ride, while offering
19 14 FR-R38R-44 FR-R38R LL Road Frame - Red, 44 LL Road Frame The LL Frame provides a safe comfortable ride, while offering
20 14 FR-R38R-48 FR-R38R LL Road Frame - Red, 48 LL Road Frame The LL Frame provides a safe comfortable ride, while offering
Query Settings
PROPERTIES
Name
AdventureWorks Product Lookup
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Added Custom
Reordered Columns
12 COLUMNS, 293 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 11:54

- Replace zeros (0) in the Product Style column with “NA”

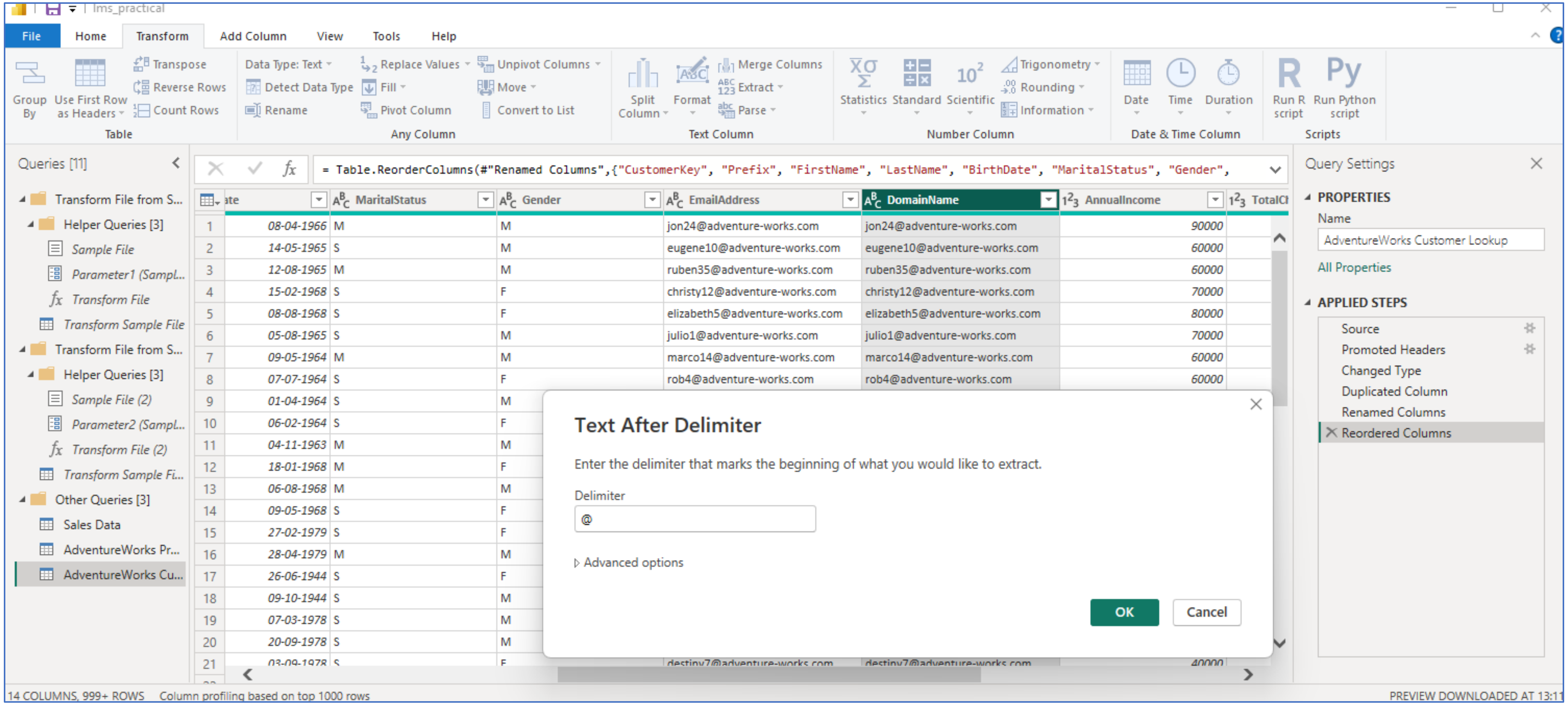


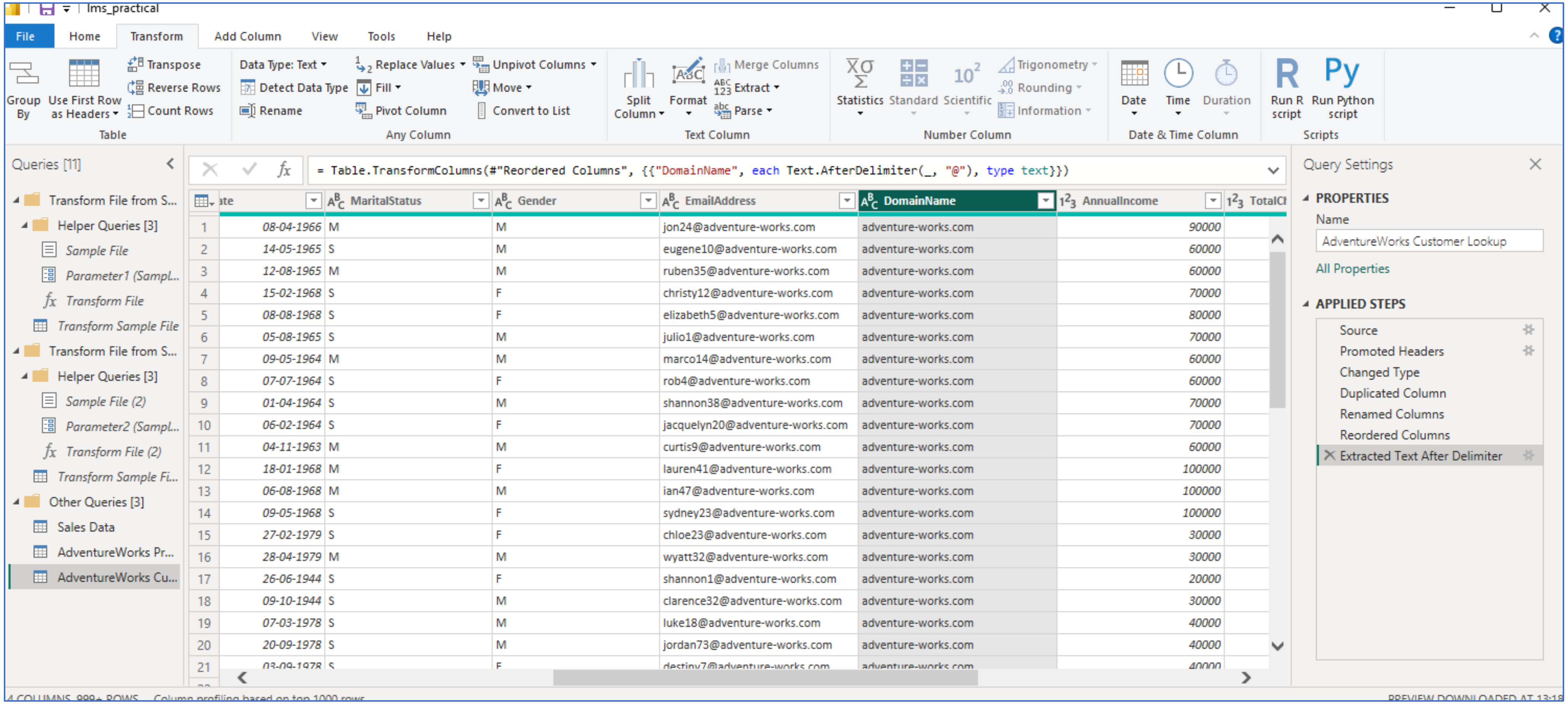
a) Implement text transformations

- Duplicate the email address column and name it “Domain Name”

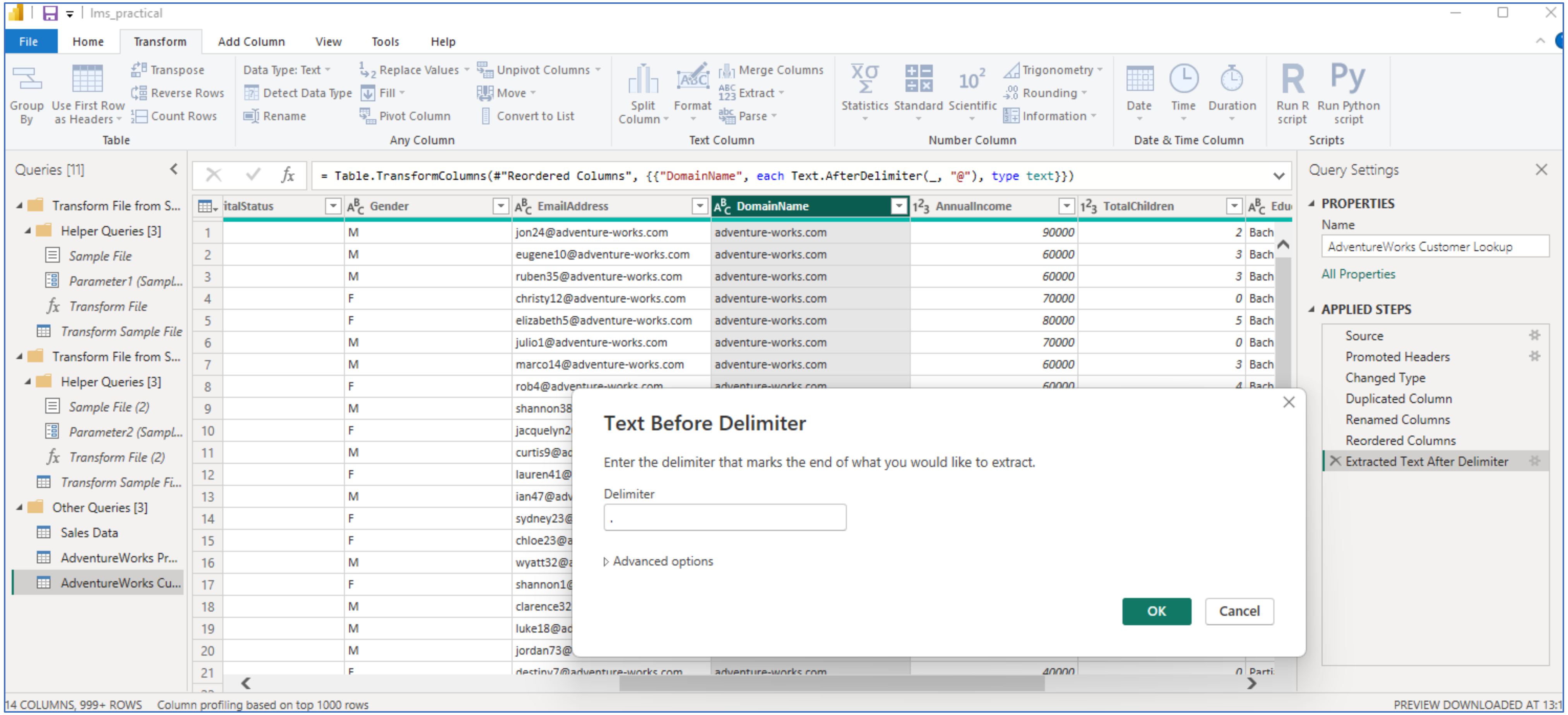


- In the new column, remove all text/characters except for the domain name

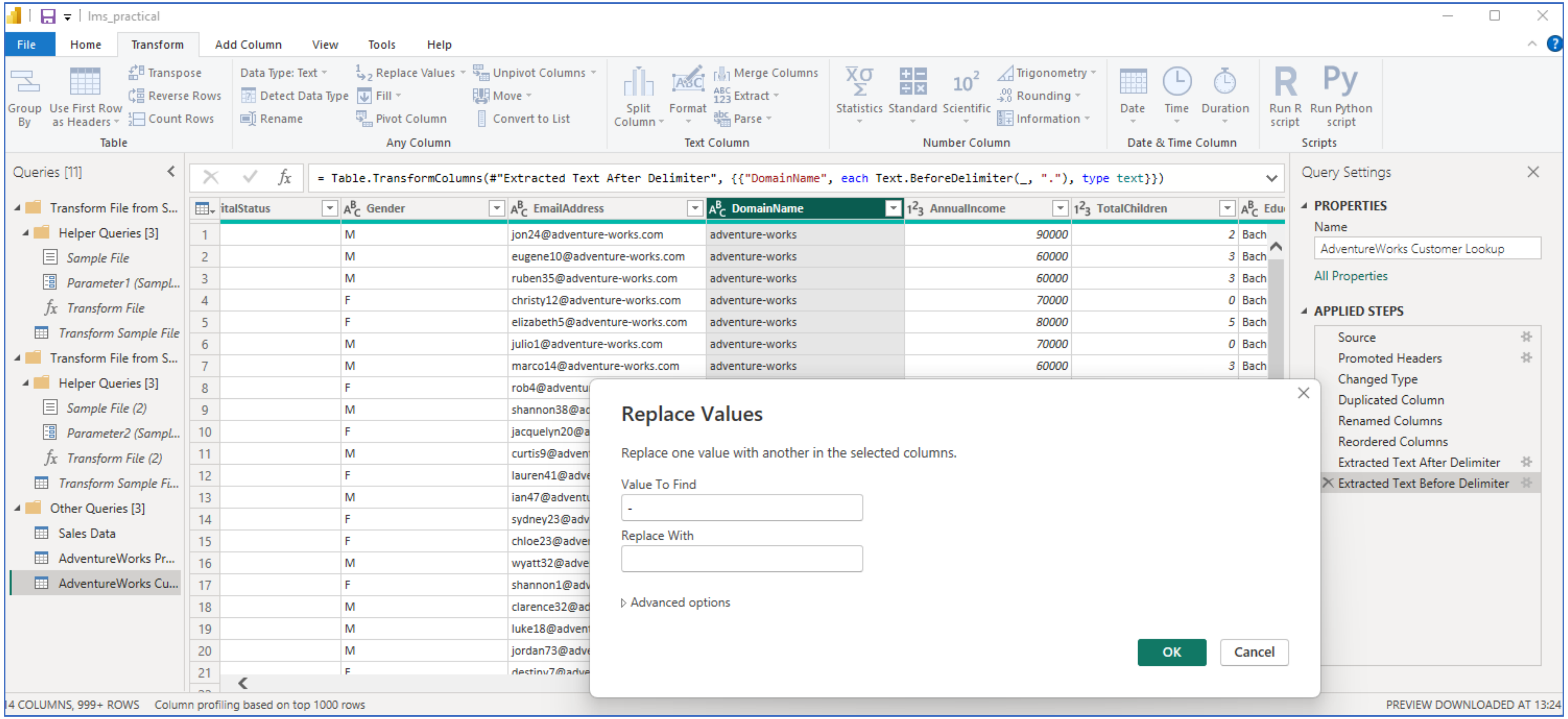



lms_practical
File
Home
Transform
Add Column
View
Tools
Help
Group By
Use First Row as Headers
Transpose
Reverse Rows
Count Rows
Table
Data Type: Text
Detect Data Type
Rename
Replace Values
Fill
Pivot Column
Unpivot Columns
Move
Convert to List
Any Column
Split Column
Format
Merge Columns
Extract
Parse
Text Column
Statistics
Standard
Scientific
Trigonometry
Rounding
Information
Number Column
Date
Time
Duration
Date & Time Column
Run R script
Run Python script
Scripts
Queries [11]
Transform File from S...
Helper Queries [3]
Sample File
Parameter1 (Sampl...
Transform File
Transform Sample File
Transform File from S...
Helper Queries [3]
Sample File (2)
Parameter2 (Sampl...
Transform File (2)
Transform Sample Fi...
Other Queries [3]
Sales Data
AdventureWorks Pr...
AdventureWorks Cu...
= Table.TransformColumns(#"Reordered Columns", {{"DomainName", each Text.AfterDelimiter(_, "@"), type text}})
ate | MaritalStatus | Gender | EmailAddress | DomainName | AnnualIncome | TotalCl
1 | 08-04-1966 | M | M | jon24@adventure-works.com | adventure-works.com | 90000
2 | 14-05-1965 | S | M | eugene10@adventure-works.com | adventure-works.com | 60000
3 | 12-08-1965 | M | M | ruben35@adventure-works.com | adventure-works.com | 60000
4 | 15-02-1968 | S | F | christy12@adventure-works.com | adventure-works.com | 70000
5 | 08-08-1968 | S | F | elizabeth5@adventure-works.com | adventure-works.com | 80000
6 | 05-08-1965 | S | M | julio1@adventure-works.com | adventure-works.com | 70000
7 | 09-05-1964 | M | M | marco14@adventure-works.com | adventure-works.com | 60000
8 | 07-07-1964 | S | F | rob4@adventure-works.com | adventure-works.com | 60000
9 | 01-04-1964 | S | M | shannon38@adventure-works.com | adventure-works.com | 70000
10 | 06-02-1964 | S | F | jacquelyn20@adventure-works.com | adventure-works.com | 70000
11 | 04-11-1963 | M | M | curtis9@adventure-works.com | adventure-works.com | 60000
12 | 18-01-1968 | M | F | lauren41@adventure-works.com | adventure-works.com | 100000
13 | 06-08-1968 | M | M | ian47@adventure-works.com | adventure-works.com | 100000
14 | 09-05-1968 | S | F | sydney23@adventure-works.com | adventure-works.com | 100000
15 | 27-02-1979 | S | F | chloe23@adventure-works.com | adventure-works.com | 30000
16 | 28-04-1979 | M | M | wyatt32@adventure-works.com | adventure-works.com | 30000
17 | 26-06-1944 | S | F | shannon1@adventure-works.com | adventure-works.com | 20000
18 | 09-10-1944 | S | M | clarence32@adventure-works.com | adventure-works.com | 30000
19 | 07-03-1978 | S | M | luke18@adventure-works.com | adventure-works.com | 40000
20 | 20-09-1978 | S | M | jordan73@adventure-works.com | adventure-works.com | 40000
21 | 03-09-1978 | S | F | destiny7@adventure-works.com | adventure-works.com | 40000
Query Settings
PROPERTIES
Name
AdventureWorks Customer Lookup
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Duplicated Column
Renamed Columns
Reordered Columns
Extracted Text After Delimiter

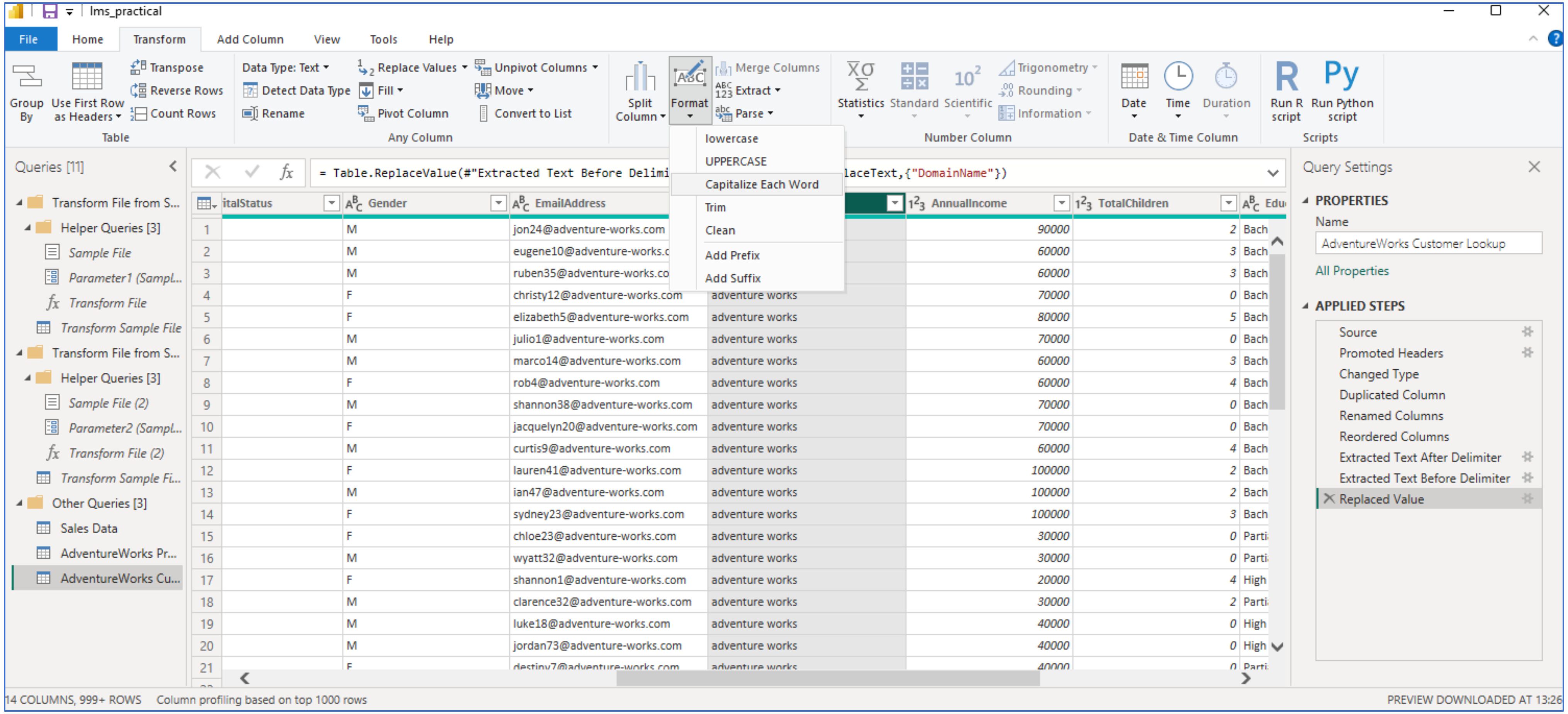
- Removing .com from domain name



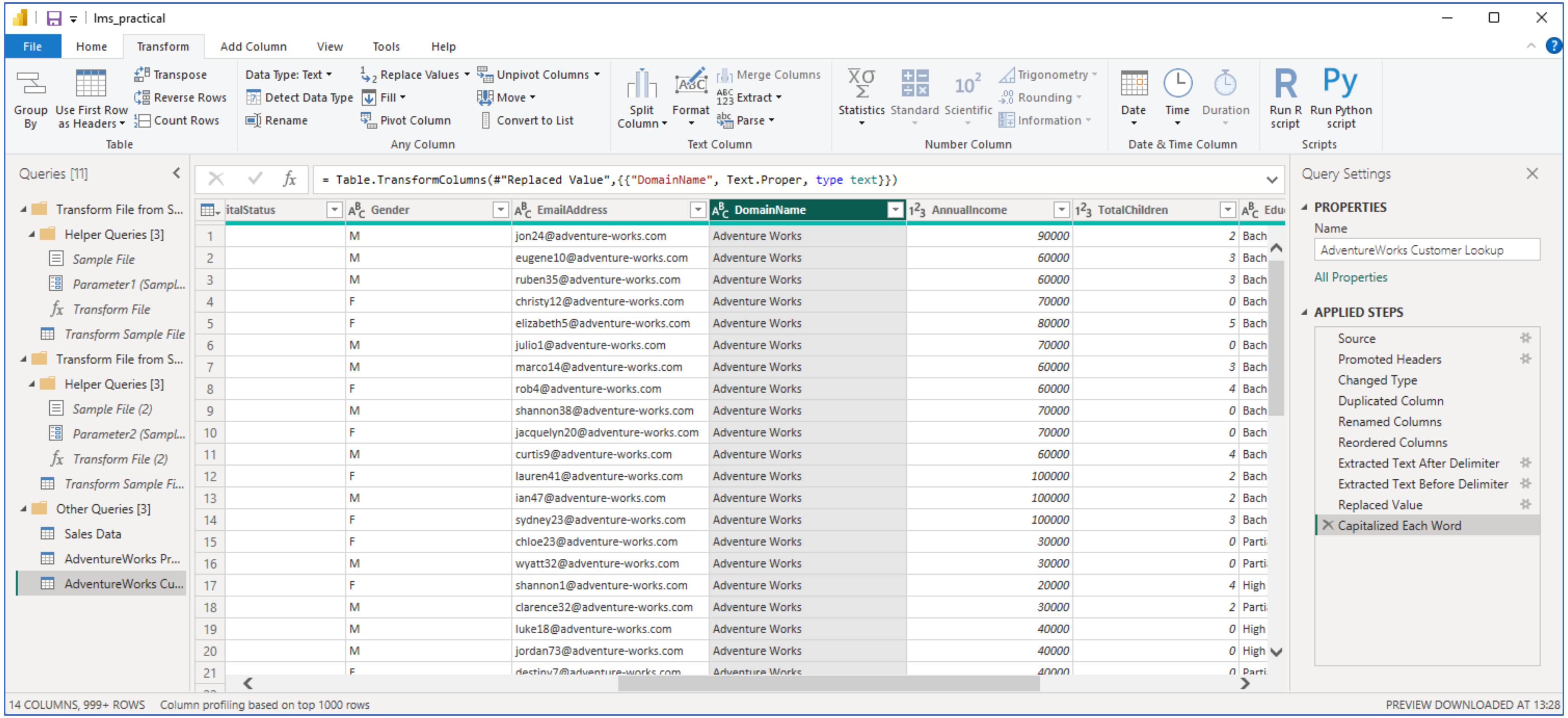
- Replacing – with ' ' (space)



- Apply formatting to capitalize each word



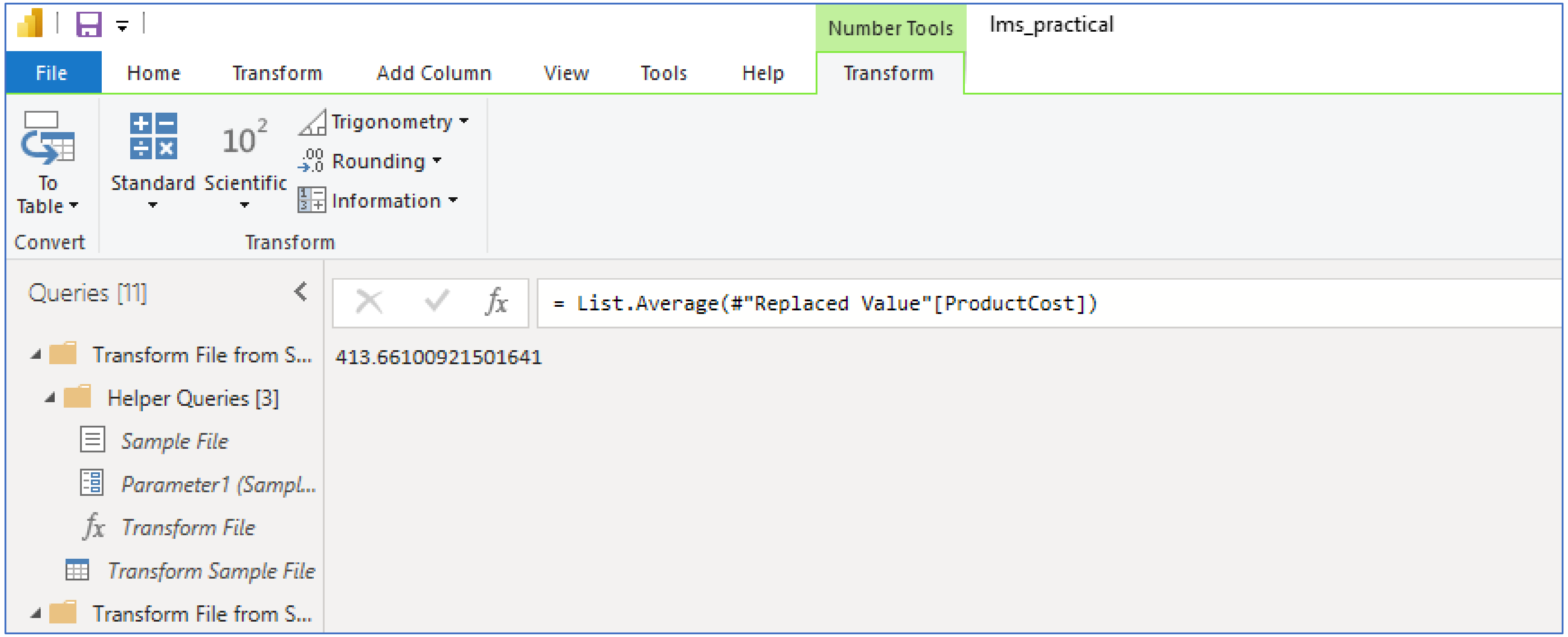
- Final Domain Name column



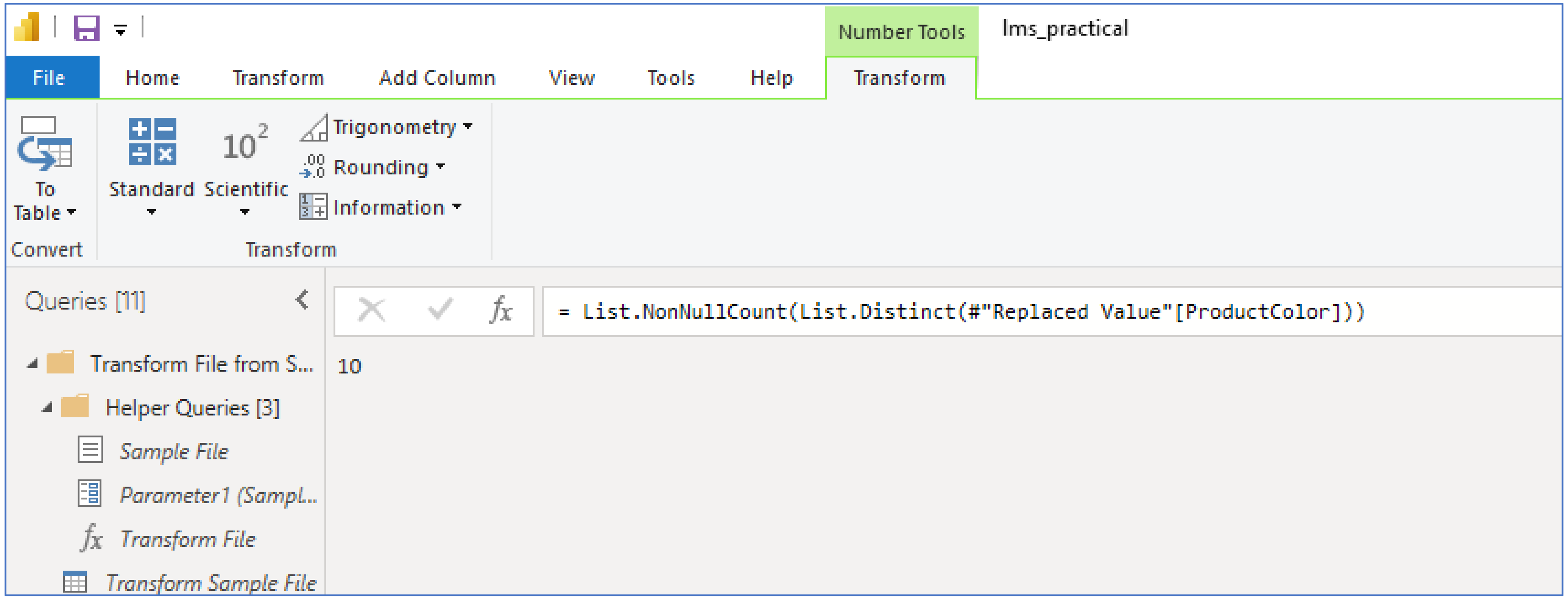
	italStatus	Gender	EmailAddress	DomainName	AnnualIncome	TotalChildren	Edu
1		M	jon24@adventure-works.com	Adventure Works	90000	2	Bach
2		M	eugene10@adventure-works.com	Adventure Works	60000	3	Bach
3		M	ruben35@adventure-works.com	Adventure Works	60000	3	Bach
4		F	christy12@adventure-works.com	Adventure Works	70000	0	Bach
5		F	elizabeth5@adventure-works.com	Adventure Works	80000	5	Bach
6		M	julio1@adventure-works.com	Adventure Works	70000	0	Bach
7		M	marco14@adventure-works.com	Adventure Works	60000	3	Bach
8		F	rob4@adventure-works.com	Adventure Works	60000	4	Bach
9		M	shannon38@adventure-works.com	Adventure Works	70000	0	Bach
10		F	jacquelyn20@adventure-works.com	Adventure Works	70000	0	Bach
11		M	curtis9@adventure-works.com	Adventure Works	60000	4	Bach
12		F	lauren41@adventure-works.com	Adventure Works	100000	2	Bach
13		M	ian47@adventure-works.com	Adventure Works	100000	2	Bach
14		F	sydney23@adventure-works.com	Adventure Works	100000	3	Bach
15		F	chloe23@adventure-works.com	Adventure Works	30000	0	Parti
16		M	wyatt32@adventure-works.com	Adventure Works	30000	0	Parti
17		F	shannon1@adventure-works.com	Adventure Works	20000	4	High
18		M	clarence32@adventure-works.com	Adventure Works	30000	2	Parti
19		M	luke18@adventure-works.com	Adventure Works	40000	0	High
20		M	jordan73@adventure-works.com	Adventure Works	40000	0	High
21		F	destiny7@adventure-works.com	Adventure Works	40000	0	Parti

b) Numeric tools

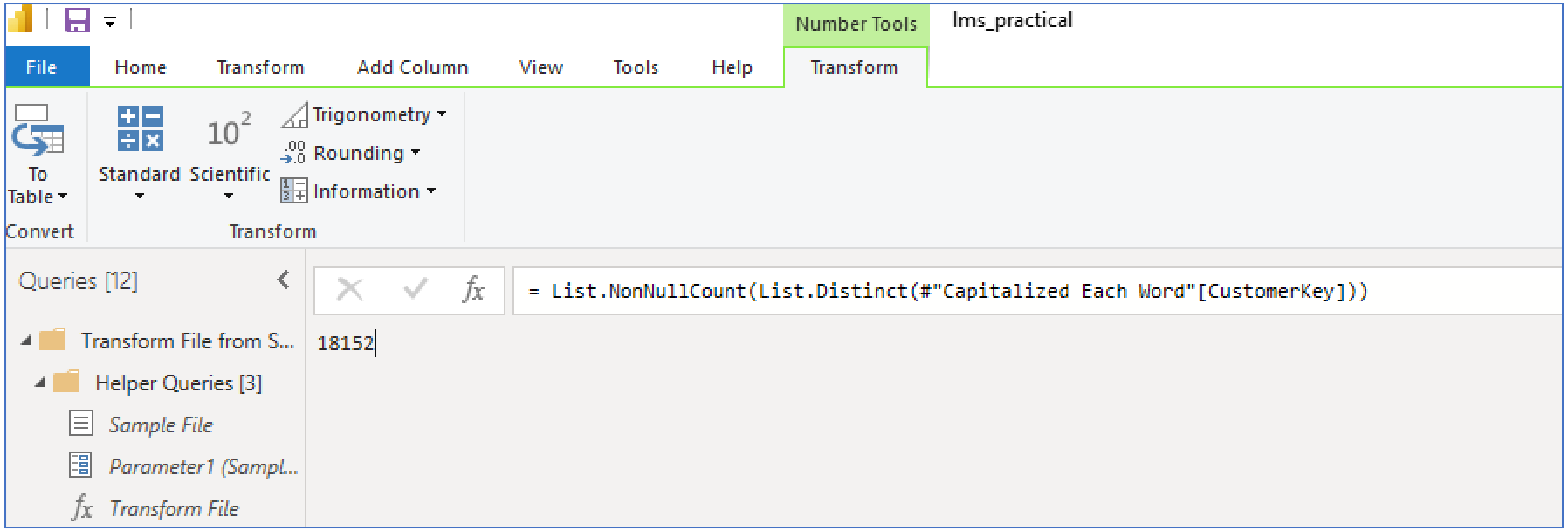
- What is our average product cost?



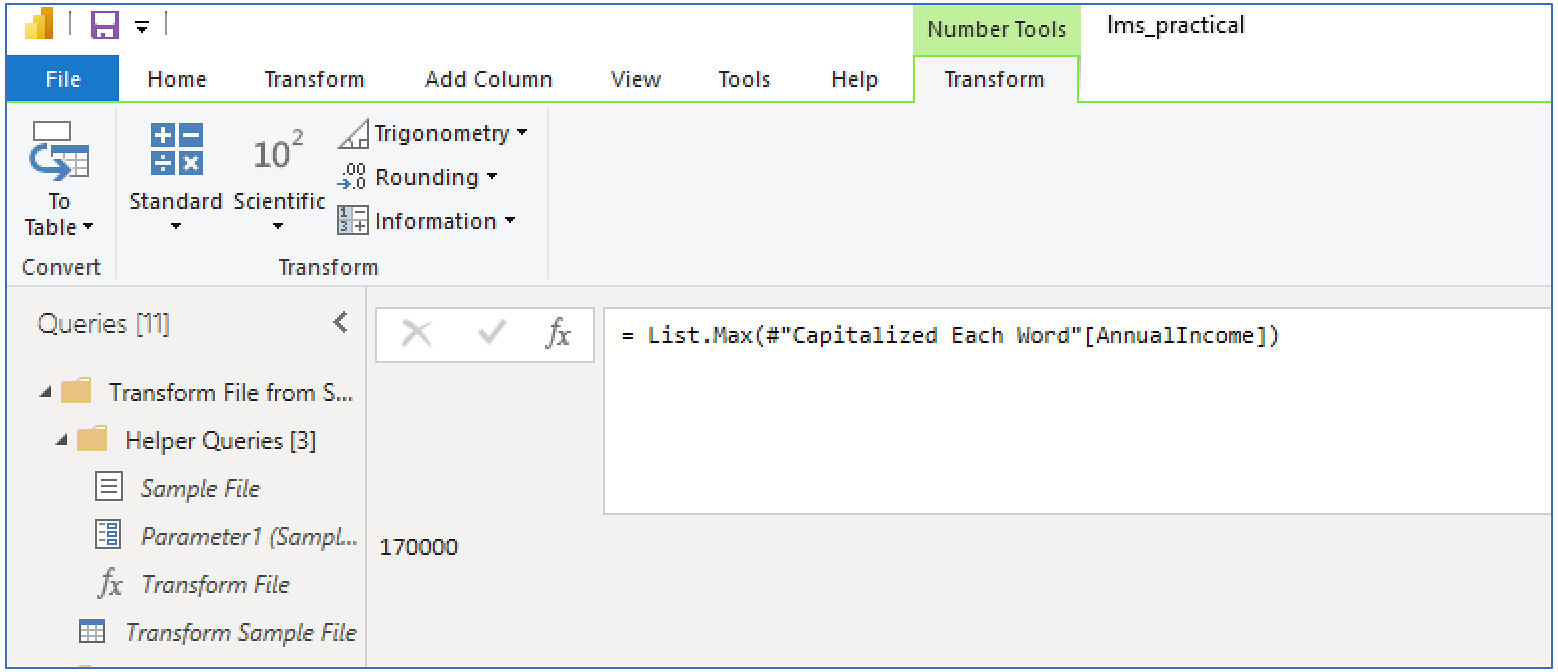
- How many colors do we sell our products in?



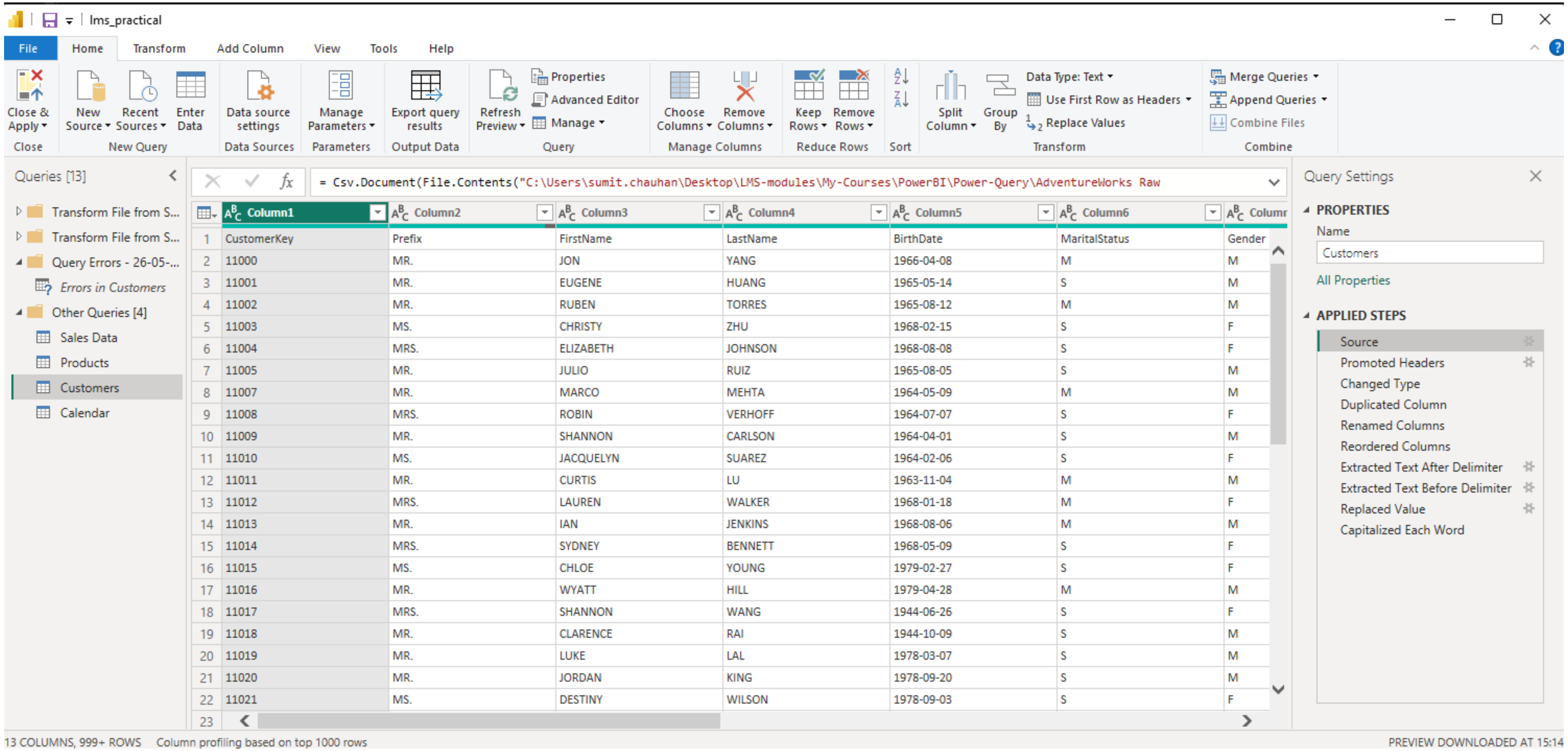
- How many distinct customers do we have?



- What is the maximum annual customer income?

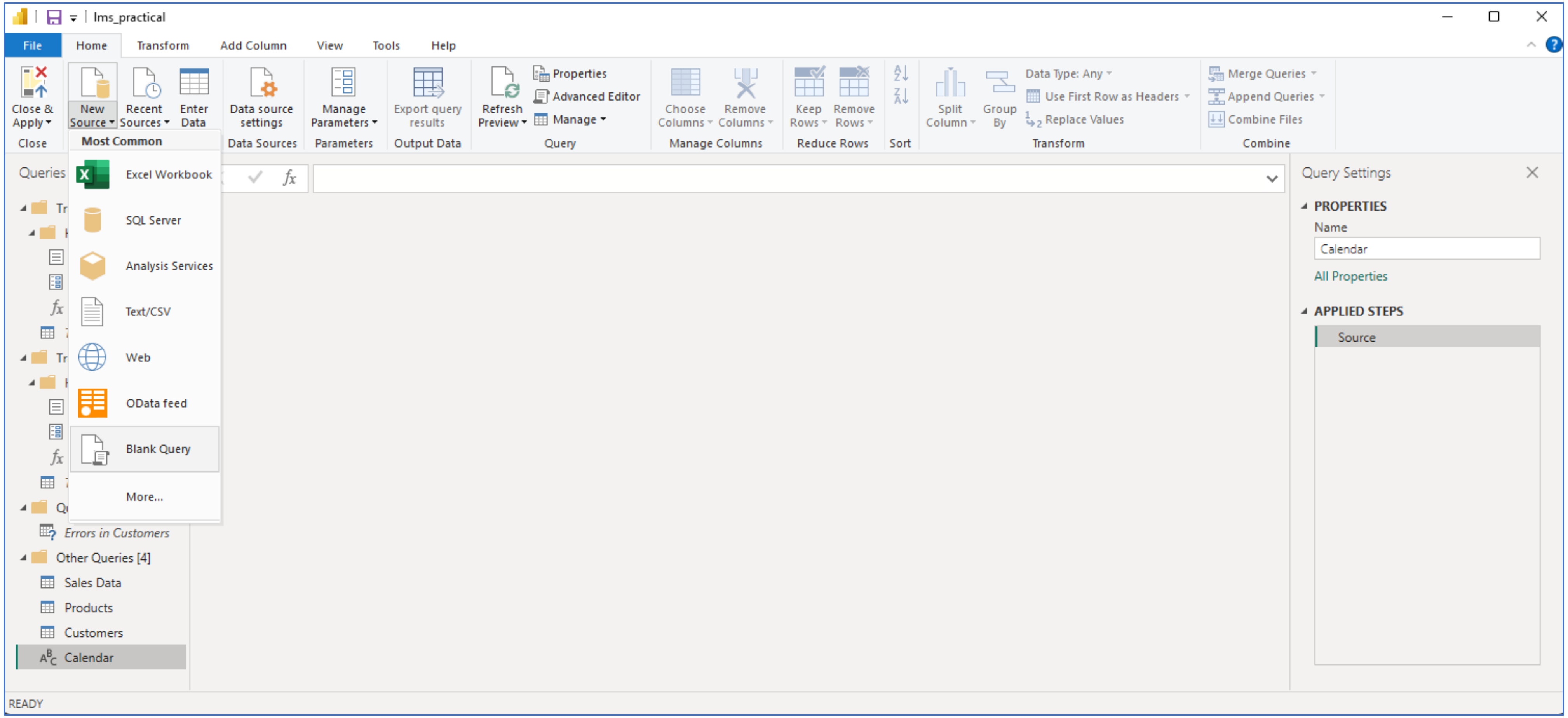


- Return the tables to their original state

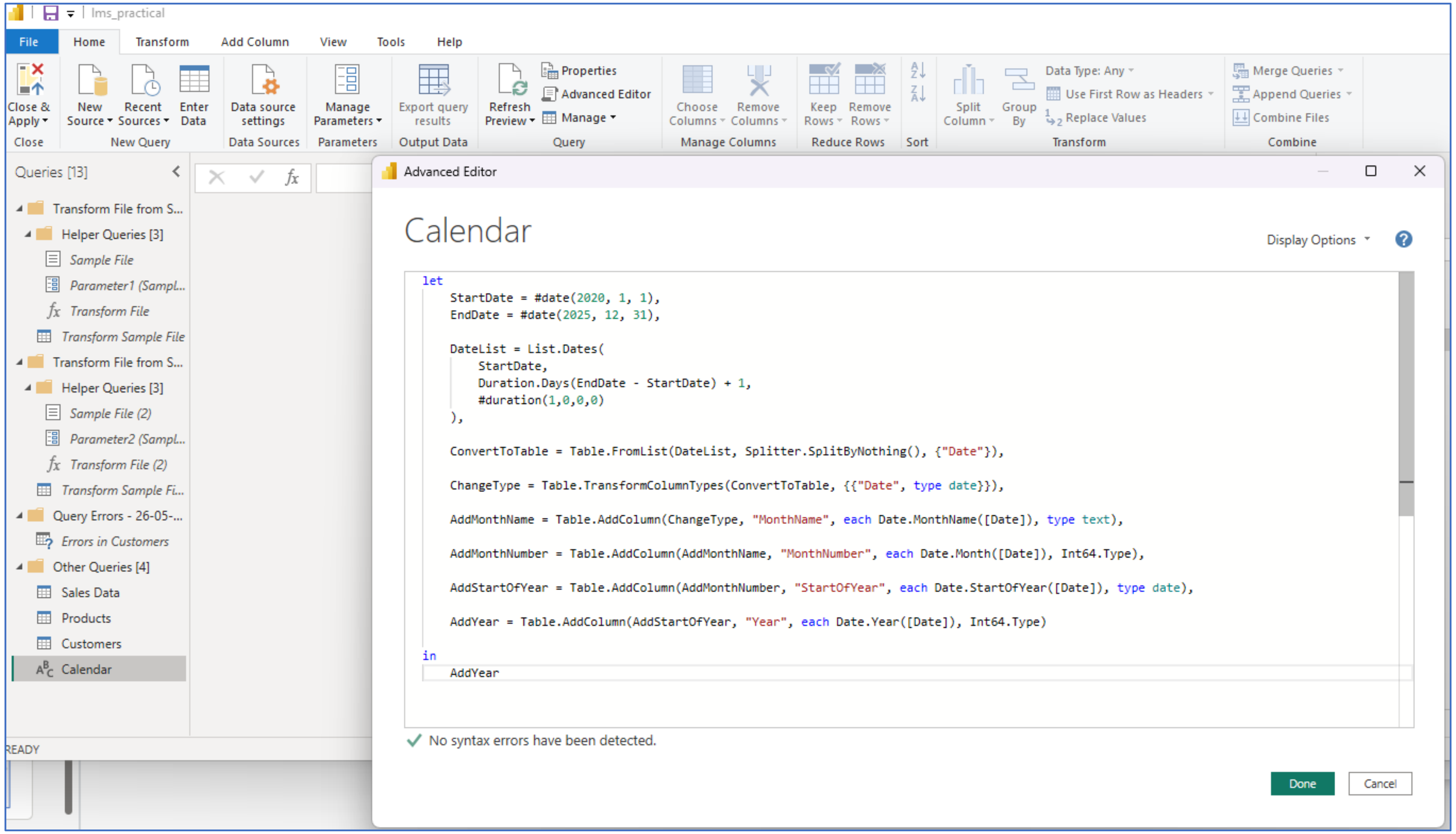


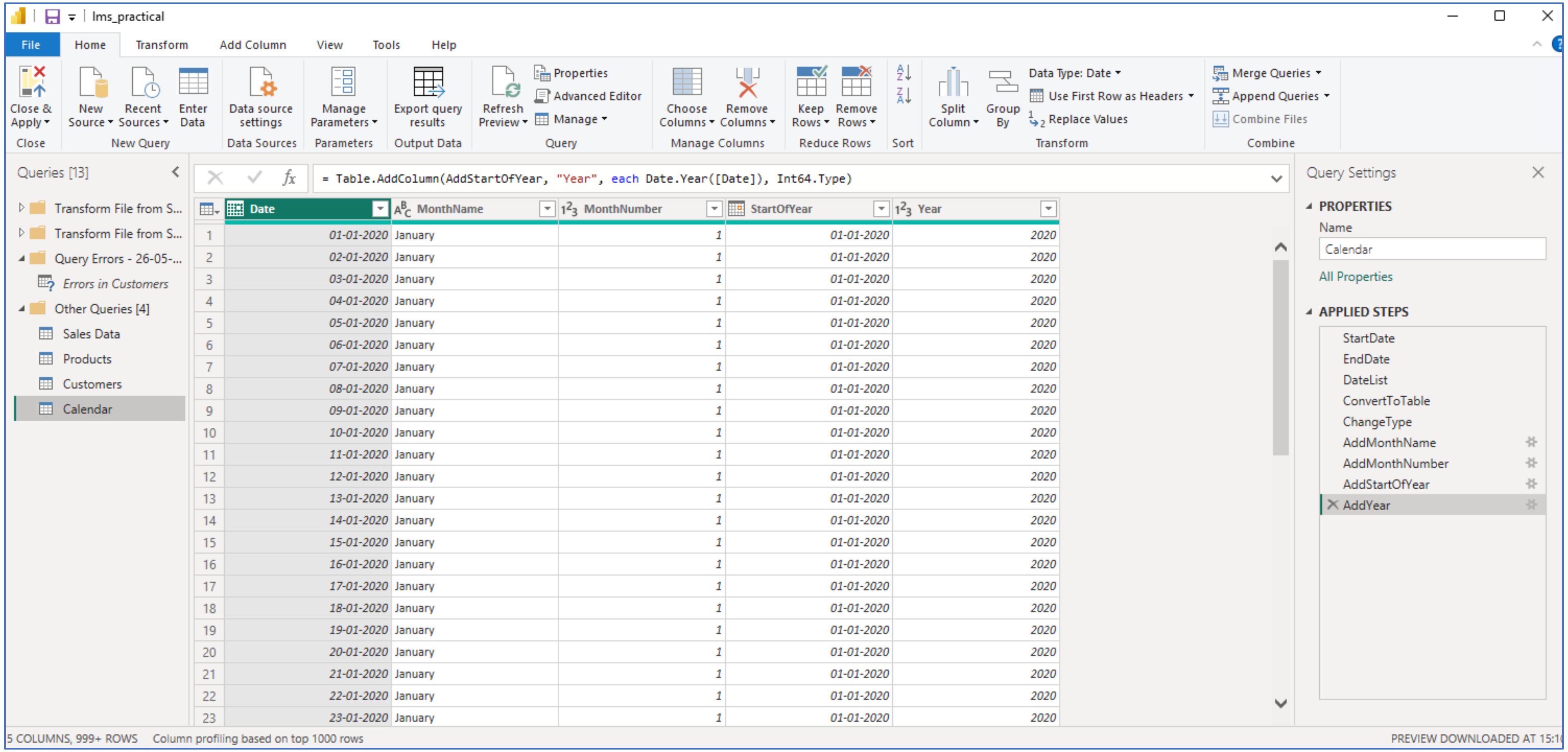
c) Date Transformations

- Create a calendar table using M language (creating blank query for it)



- Advanced editor




lms_practical
File
Home
Transform
Add Column
View
Tools
Help
Close & Apply
New Source
Recent Sources
Enter Data
Data source settings
Manage Parameters
Export query results
Refresh Preview
Properties
Advanced Editor
Manage
Choose Columns
Remove Columns
Keep Rows
Remove Rows
Split Column
Group By
Data Type: Date
Use First Row as Headers
Replace Values
Merge Queries
Append Queries
Combine Files
Close
New Query
Data Sources
Parameters
Output Data
Query
Manage Columns
Reduce Rows
Sort
Transform
Combine
Queries [13]
Transform File from S...
Transform File from S...
Query Errors - 26-05-...
Errors in Customers
Other Queries [4]
Sales Data
Products
Customers
Calendar
= Table.AddColumn(AddStartOfYear, "Year", each Date.Year([Date]), Int64.Type)
Date
MonthName
MonthNumber
StartOfYear
Year
01-01-2020 January 1 01-01-2020 2020
02-01-2020 January 1 01-01-2020 2020
03-01-2020 January 1 01-01-2020 2020
04-01-2020 January 1 01-01-2020 2020
05-01-2020 January 1 01-01-2020 2020
06-01-2020 January 1 01-01-2020 2020
07-01-2020 January 1 01-01-2020 2020
08-01-2020 January 1 01-01-2020 2020
09-01-2020 January 1 01-01-2020 2020
10-01-2020 January 1 01-01-2020 2020
11-01-2020 January 1 01-01-2020 2020
12-01-2020 January 1 01-01-2020 2020
13-01-2020 January 1 01-01-2020 2020
14-01-2020 January 1 01-01-2020 2020
15-01-2020 January 1 01-01-2020 2020
16-01-2020 January 1 01-01-2020 2020
17-01-2020 January 1 01-01-2020 2020
18-01-2020 January 1 01-01-2020 2020
19-01-2020 January 1 01-01-2020 2020
20-01-2020 January 1 01-01-2020 2020
21-01-2020 January 1 01-01-2020 2020
22-01-2020 January 1 01-01-2020 2020
23-01-2020 January 1 01-01-2020 2020
Query Settings
PROPERTIES
Name
Calendar
All Properties
APPLIED STEPS
StartDate
EndDate
DateList
ConvertToTable
ChangeType
AddMonthName
AddMonthNumber
AddStartOfYear
AddYear
5 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 15:1